```
GiE          F1          0          VALUE={v(ie)*g}
*            voltage components
EuxS   usxS   0   VALUE={(2*v(u1,u2)-v(v1,v2)-v(w1,w2))/3}
EuyS   usyS   0   VALUE={(v(v1,v2)-v(w1,w2))/sqrt(3)}
Eux    usx    0   VALUE={ v(usxS)*cos(v(alpha))+v(usyS)*sin(v(alpha))}
Euy    usy    0   VALUE={-v(usxS)*sin(v(alpha))+v(usyS)*cos(v(alpha))}
* Starpoint
R2    v2    GND1   1u
R12   u2    GND1   1u
R23   w2    GND1   1u
RGND1  GND1  0  1e6
*            Zerosystem
Eu0  u_Null 0  VALUE={(v(u1,u2)+v(v1,v2)+v(w1,w2))/3}
Ei0   i_Null  0  VALUE={idt(v(u_Null)-R0*v(i_null))/L0}
*            current components
EisxS  isxS  0 VALUE={v(isx)*cos(v(alpha))-v(isy)*sin(v(alpha))}
EisyS  isyS  0 VALUE={v(isy)*cos(v(alpha))+v(isx)*sin(v(alpha))}
*            magnetic coupling of direct axis with saturation
Gpsix  0  isx VALUE={v(psisx)}
Rsx    isx  0  {1/Lsd}
Gpsidx isx iDx1 VALUE={v(psiDx)}  ;
Rsdx   iDx1 isx  {1/LDx}
GpsiEx iDx1 imx1 VALUE={v(psiE)}
RsEx   imx1 iDx1 {1/LE}
vphx    imx1 imxa1 0
Eimx   imxa1  0  VALUE={ImNx*(a1*i(vphx)/PsiN+(1-a1)*PWRS(i(vphx)/PsiN,m1))}
*            magnetic coupling of quadrature axis with saturation
Gpsiy   0  isy  VALUE={v(psisy)}
Rsy     isy   0  {1/Lsq}
Gpsidy  isy   imy1  VALUE={v(psiDy)}
Rsdy    imy1  isy  {1/LDy}
vphy    imy1   imya1  0
Eimy    imya1  0  VALUE={ImNy*(a2*i(vphy)/PsiN+(1-a2)*PWRS(i(vphy)/PsiN,m2))}
*            damper currents
Eidx  idx 0 VALUE={v(idx1)-v(isx)}
Eidy  idy 0 VALUE={v(imy1)-v(isy)}
*            exciter current
Eie ie 0 VALUE={v(imx1)-v(idx1)}
*            Flux components
Epsi_sx  psisx   0 VALUE={err0*PsiN+idt(v(usx)-Rs*v(isx)+p*(2*pi)/60*v(n)*v(psisy))}
Epsi_sy  psisy   0 VALUE={idt(v(usy)-Rs*v(isy)-p*(2*pi)/60*v(n)*v(psisx))}
Epsi_Dx  psiDx  0 VALUE={err0*psiN+idt(-RDx*v(idx))}
Epsi_Dy  psiDy  0 VALUE={idt(-RDy*v(idy))}
Epsi_E   psiE  0 VALUE={err0*psiN*(1+Streuung)+idt(v(F1,0)*g*RE_factor-RE*RE_factor*g*g*v(ie))}
*            Load angle
Etheta 0 theta VALUE={idt(180/pi*atan2(v(usx),v(usy)))}
*            inner torque
Emi  Mi  0 VALUE={1.5*p*(i(vphx)*v(isy)-i(vphy)*v(isx))}
*            outher torque
EMoment  M 0 VALUE={-v(Mi)+v(Rbg)}
Vn n 0 {n1}
* friction
GRbg   0  Rbg  TABLE {v(n)} (-1,-1) (1,1)
RRbg   Rbg   0  {MN*rbg}
* Compute angle alpha
Ealpha alpha  0  VALUE={alpha_0*pi/180+idt(v(n))*p/60*(2*pi)}
.ENDS
```